J SYST SCI SYST ENG
Vol. 34, No. 4, August 2025, pp. 471–489
DOI: https://doi.org/10.1007/s11518-025-5651-2
ISSN: 1004-3756 (paper), 1861-9576 (online)
CN 11-2983/N

Examining the Relationship between Coalition Voting and Knowledge Contribution in Decentralized Autonomous Organizations

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Abstract. Decentralized Autonomous Organizations (DAOs) leverage blockchain technology to facilitate community governance and incentivize users to contribute more to the community through a fair distribution of rewards. However, the emergence of coalition voting–where groups of users collaborate to secure higher token rewards or other advantages–poses a double-edged sword. On one hand, coalition voting can compromise the fairness and integrity of the voting process. On the other hand, it may enhance user interactions, promote deeper collaboration, and facilitate the exchange of information, potentially leading to increased knowledge contributions within the community. This dual nature creates ambiguity regarding the overall impact of coalition voting on knowledge sharing in DAOs. Utilizing data from Steemit, this study employs complex network analysis and Panel Vector Autoregression (PVAR) models to investigate the interplay between coalition voting and user knowledge contributions. Theoretically, this research advances the knowledge management literature by highlighting the nuanced role of coalition voting in fostering user engagement despite its governance-related drawbacks. Practically, it offers valuable insights for DAO communities in developing effective monitoring systems and governance strategies that harmonize incentive structures with equitable community participation.

Keywords: Knowledge contribution, token incentives, coalition voting, blockchain

History: Received: June 7, 2024; Revised: December 11, 2024; Accepted: February 5, 2025; Published online: March 31, 2025

* The main contents of this work were previously presented at the 22nd International Symposium on Knowledge and Systems Sciences (KSS2023) held in Guangzhou during December 2-3, 2023.

1. Introduction

Knowledge sharing, pivotal in knowledge management (Wasko and Faraj 2005), entails the knowledge flow between individuals. It encompasses knowledge contribution and knowledge seeking (Chen and Hung 2010). The former involves providing valuable content for the community, while the latter facilitates users to search, browser, and disseminate knowledge content within the platform (Bock et al. 2005). The success of a community hinges on knowledge contribution, only when users proactively partake in knowledge contribution can a community showcase rich knowledge content and sustain long-term vitality (Chen et al. 2019). However, not all participants in the knowledge community contribute to knowledge, with the majority post less (Baym 2000). Many users tend to free ride or make insuffi-

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cient contributions, precipitating the tragedy of public resources (Hardin 2009). Furthermore, although users make significant contributions to the community through knowledge sharing, these contributors often do not receive corresponding economic compensation, resulting in user attrition from the platform (Chen et al. 2019). Consequently, motivating users to contribute continuously is a formidable challenge for the sustainable development of online knowledge communities (Wang and Fesenmaier 2004).

Online knowledge communities with blockchain token incentives have emerged to meet this need. These blockchain token incentive-based knowledge communities are decentralized autonomous organizations, represent a new type of knowledge sharing ecosystem that equitably and transparently distributes token awards and community privileges to all users contributing to the community's development (Scott and Larimer 2017). This innovative incentive mechanism strives to address the incompatibility of knowledge contribution incentives in traditional online knowledge communities by surmounting the free riding problem and ensuring contributors receive commensurate economic compensation through blockchain token incentives. Taking Steemit, the most representative blockchain-based knowledge community worldwide, as an example, by January 2020, its registered users reached 1.34 million, and the community reward amount for users' knowledge contribution behavior was nearly 60 million dollars.

However, the introduction of blockchain tokens into the knowledge community transforms knowledge production and discovery into investment behavior. Blockchain token-powered online knowledge communities encourage knowledge sharing behavior through token rewards. Different communities share virtually the same design and incentive mechanism: individual users vote on content quality, which is weighted according to all voting users' token holdings, and rewards are distributed among users in proportion to their token holdings (Tsoukalas and Falk 2020). Under this mechanism, users may engage in collusion. One form of collusion involves bribing users with a large number of tokens to secure their votes (Li and Palanisamy 2019). Li and Palanisamy (2019) have discovered such collusion on the blockchain by constructing complex networks with transfer data. Another collusion form, like a Sybil attack, involves multiple users uniting to vote with the aim of achieving common goals or interests, thereby earning higher token rewards or other benefits(Sun et al. 2022).

Although existing studies have extensively evaluated the effectiveness of the incentive mechanisms in the Steemit community, their findings show significant discrepancies and have yet to reach a unified conclusion (Li and Palanisamy 2019, Tsoukalas and Falk 2020). Moreover, some scholars have explored the impact of individual voting behavior on users' continuous knowledge contribution in traditional online communities, finding that voting behavior can promote knowledge sharing and community vitality to some extent (Chen et al. 2019, Bock et al. 2005). However, in blockchain-driven Decentralized Autonomous Organizations (DAOs), voting be-

havior is not merely a simple evaluation of content; it also involves complex motivations to obtain token rewards through collective decision-making (Sun et al. 2022). Under such mechanisms, users may engage in collaborative actions to optimize their benefits, such as coordinating voting strategies to secure higher rewards or influence community decisions (Li and Palanisamy 2019). Nonetheless, existing literature has not sufficiently examined how collective voting behavior under token-based incentive mechanisms affects the depth and breadth of users' knowledge contributions, nor how these impacts evolve over time (Chen and Hung 2010). Specifically, prior research has yet to thoroughly uncover how token rewards obtained through collective voting in DAOs influence users' knowledge contribution behaviors and how such effects vary across different time scales. Therefore, this study aims to fill this research gap.

Steemit, being the earliest and largest knowledge community based on blockchain token incentives, has become scholars' primary research subject. By use data from 5764 active users from January to December 2019 on Steemit, we investigates the relationship between user coalition voting and knowledge contribution. Firstly, we identified coalition voting behavior through constructing a complex network model. In the network construction process, we discard common voting attributable to users' similar knowledge and identify voting coalition groups using the Louvain algorithm. For a specific vote, if the majority of users in the same coalition participate, we define this vote as a coalition vote. We use statistical methods to calculate a threshold for identifying coalition voting, based on deviations from expected joint voting behavior, and apply the Louvain algorithm to detect potential voting coalitions within the user network. Secondly, according to the relevant literature, we divide user knowledge contribution into post number and knowledge innovation contribution, where knowledge innovation includes the breadth and depth of the knowledge network. Finally, the relationship between knowledge contribution and coalition voting is studied by Panel Vector Autoregression (PVAR) model.

This study contributes to both research and practice. Firstly, it enriches knowledge management research. Most previous studies focused on the impact of single incentive factors (such as reputation and economic incentives) on community users' knowledge contribution, resulting in relatively simplistic findings. Secondly, this study's results indicate that user coalition voting has a certain promoting effect on users' knowledge contribution, even though coalition voting is not the behavior that communities desire from users. Thirdly, this study finds that the relationship between user coalition voting and knowledge contribution is dynamic. In practice, the user coalition voting detection algorithm developed in this study can provide a method for detecting user collusion for blockchain platforms based on token weights and offer governance decision-making support for blockchain communities based on token incentives.

The rest of the paper is organized as follows. Section 2 offers a literature review. Section 3 provides the theoretical background and develops the hypotheses for our empirical analysis. Section 4 presents the General

Introduction of Steemit and discusses Coalition Voting by Users on Posts. Section 5 details the construction of our model variables. Section 6 presents the PVAR models and discusses the estimation results. Section 7 discusses the findings, theoretical and practical implications, and concludes the paper.

2. Literature Review

2.1 User Contribution in Online Communities

Research on user contributions in online knowledge communities provides valuable insights into the factors influencing user behavior, broadly categorized into self-interest motivations and social factors. While these dimensions are well-established, the current body of work reveals several limitations that merit further exploration.

2.1.1 Self-Interest Motivations

The emphasis on self-interest, encompassing economic and psychological incentives, has illuminated the drivers of individual knowledge contributions (Bock et al. 2005, Gneezy et al. 2011). However, much of this research operates within the framework of conventional, centralized platforms. Studies like (Frey and Oberholzer-Gee 1997) and (Rocha and Ghoshal 2006) have explored intangible rewards, such as appreciation or a sense of purpose, but they often overlook how these dynamics evolve in decentralized contexts where governance mechanisms differ substantially.

2.1.2 Social and Technological Factors

Research by (Chai and Kim 2012) and (Kang et al. 2017) highlights the importance of reputation, identity, and technical usability. While these studies effectively address user interaction within centralized platforms, the transition to blockchain-based communities introduces unique challenges, such as governance without intermediaries and potential inequalities due to token-based economies. These aspects remain insufficiently explored in existing literature.

2.1.3 Token Economies and Blockchain Knowledge Communities

The advent of token economies has shifted the paradigm of user incentives. Blockchain platforms, such as Steemit, were initially lauded for dismantling centralized information barriers and enabling value monetization (Scott and Larimer 2017). However, subsequent research revealed critical issues, including fluctuating token values, user attrition, and governance centralization. Critics, such as Tsoukalas and Falk (2020), have raised concerns about the dominance of early adopters and the emergence of power centers, which undermine the principle of decentralization.

Moreover, while economic incentives may enhance knowledge contributions temporarily (Li and Palanisamy 2019), their long-term efficacy is questionable. Studies suggest that reliance on financial rewards could suppress intrinsic motivation over time, leading to diminished contributions (Kiayias et al. 2018). Yet, this potential trade-off between short-term gains and long-term community health has not been adequately quantified.

Unlike prior research, which predominantly focuses on individual user motivations or broad system-level challenges, this study highlights the dual-edged nature of coalition voting–its potential to foster collaboration and

knowledge exchange while raising governance equity concerns.

2.2 Coalition in Decentralized Autonomous Organizations

Coalition voting in blockchain-based decentralized autonomous organizations (DAOs) represents a distinctive behavior where users collaborate to achieve shared objectives, such as influencing decisions or maximizing token rewards (Sun et al. 2022). While this behavior can enhance cooperative decision-making, existing research often frames coalition voting narrowly as a governance challenge. For instance, studies like (Li and Palanisamy 2019) underscore how such alliances can undermine the effectiveness of consensus mechanisms, such as Delegated Proof of Stake (DPOS), by enabling miners or influential users to manipulate delegate selection processes. However, this focus on governance efficiency often neglects the potential positive impacts of coalition voting, such as fostering collaboration and enhancing knowledge sharing within DAO communities.

Moreover, current research primarily identifies coalition voting through empirical methods, such as social network analysis (Li and Palanisamy 2019), without adequately exploring its broader implications for user engagement and knowledge contributions. While alliance behaviors are recognized as a form of collective action unique to decentralized platforms, studies tend to dichotomize their effects–viewing them either as harmful governance disruptions or as neutral phenomena. This binary perspective fails to capture the nuanced trade-offs between coalition votings governance challenges and its community-building potential. Additionally, the literature largely overlooks ordinary users motivations and the long-term effects of coalition behaviors on token economies and platform sustainability, focusing instead on miners or influential users.

Unlike prior studies that emphasize governance inefficiencies, this research highlights the dual-edged nature of coalition voting, exploring its potential to simultaneously undermine governance structures and enhance knowledge contributions. By applying complex network analysis and Panel Vector Autoregression (PVAR) models, it provides a comprehensive, data-driven understanding of coalition behaviors over time, addressing gaps in theoretical and empirical analyses.

3. Hypotheses Development

3.1 User Characteristics and Coalition Voting

Coalition voting refers to the act of multiple users uniting to vote with the aim of achieving common goals or interests. This behavior is particularly prevalent in blockchain and decentralized autonomous organizations (DAOs), where users coordinate votes to influence decisions and resource allocation, thereby earning higher token rewards or other benefits (Sun et al. 2022). As a group behavior rather than an individual one, coalition voting has significant social attributes. Effective coalition voting requires coordination and cooperation among users within the coalition, which are closely related to individual user characteristics. According to social capital theory (Swan-

son et al. 2020), factors influencing coalition behavior can be categorized into network structure, trust, and reciprocity, all of which can be reflected in a user's number of followers, followings, the user's tenure in the community.

In communities, users with a high number of followers often hold significant centrality and influence within the network. Through interactions with numerous followers, they accumulate a high degree of trust and social capital. This trust and influence enable them to engage followers to participate in coalition voting to achieve common goals (Sun et al. 2022). In social capital theory, the number of followers reflects the user's important position in the network structure and the trust they receive from the community (Jang et al. 2024). Therefore, we hypothesize

Hypothesis 1 *Users with a higher number of followers are more likely to participate in coalition voting.*

Users with a large number of followings typically have extensive connections and interactions within the community. This network of relationships enhances their social capital, enabling them to more effectively engage in resource exchange and reciprocal behavior Pan et al. (2015), Wang et al. (2024).Therefore, we hypothesize

Hypothesis 2 *Users with a higher number of followings are more likely to participate in coalition voting.*

Users who have been members of the community for a shorter duration are more eager to establish their reputation and social capital within the community Knoben (2008), thereby gaining more resources and opportunities in future interactions. They may wish to quickly integrate into the community and gain recognition and support from other members through active participation and organizing coalition votes.

Hypothesis 3 *Users with a shorter duration of community membership are more likely to participate in coalition voting.*

3.2 Coalition Voting and Knowledge Contribution

Users' knowledge sharing in communities is influenced not only by individual characteristics but also by various factors such as organizational atmosphere, personal traits, attitudes, organizational motivation, and trust. These factors can be divided into environmental and personal factors (Camilleri and Kozak 2022). Previous studies have indicated that, apart from personal expertise, reputation, and duration of community membership (Hsu et al. 2007), social capital such as structural capital, cognitive capital, and relational capital also impacts users' knowledge contribution in communities (Bock et al. 2005).

Users participating in coalition voting increase their social capital within the coalition in addition to their existing followers and followings in the community. Research has shown that social interactions in communities promote users' continuous contributions (Jabr 2014). Users' posting behavior is influenced by the posting behavior of their followers and those they follow (Burke et al. 2010). When users receive more feedback from their posts in the community, they are more likely to continue posting (Guan et al. 2018). Participating in coalition voting undoubtedly leads to more

votes and feedback, thereby promoting continuous contribution in the community. Therefore, we hypothesize:

Hypothesis 4 *Participation in coalition voting increases the number of posts by users in the community.*

In our study, we introduce knowledge networks to measure users' innovativeness, including the breadth and depth of their knowledge networks. Knowledge network breadth represents the diversity of users' posts, while knowledge network depth represents the specialization of users' posts. In social network theory, weak ties and strong ties are important concepts (Granovetter 1973). Here, we consider coalition voting to be closer to weak ties. Coalition voting relies on extensive mobilization, information dissemination, and strategic temporary cooperation. By participating in coalition voting, users can influence decisions and resource allocation to earn higher token rewards (Sun et al. 2022). Weak ties are more likely to introduce new knowledge and disseminate new content (Campbell et al. 1986, Levin and Cross 2004). Members of coalition voting may come from different knowledge domains within the community, and these weak ties help connect different knowledge fields and specialties, promoting users' knowledge innovation. Therefore, we hypothesize:

Hypothesis 5 *Participation in coalition voting increases the breadth of users' knowledge networks of their posts.*

Hypothesis 6 *Participation in coalition voting increases the depth of users' knowledge networks of their posts.*

4. Research Context

4.1 General Introduction of Steemit

Steemit is a decentralized content-sharing platform based on the Steem blockchain. Users can post original content and engage with others through voting, comments, and interactions. The reward system of Steemit relies on three types of tokens: STEEM, Steem Power (SP), and Steem Dollar (SBD). STEEM is a cryptocurrency that can be traded on external exchanges, and its value is determined by market supply and demand. Steem Power (SP) represents the voting weight on the platform. Holders increase their voting influence by accumulating SP, thus earning more rewards based on their voting power. Steem Dollar (SBD) is a stablecoin pegged to the U.S. dollar, and holders can redeem it for fiat currency, typically used for content creation rewards on the platform.

Steemit generates new tokens daily through blockchain technology and adds them to the reward pool, which is then distributed to content creators and voters based on their participation. Content exposure and rewards are determined by user votes, and the weight of each vote is based on the user's SP. SP not only affects voting weight but also determines the distribution of post rewards. Users accumulate SP through activities such as posting content, upvoting, commenting, and holding STEEM, reflecting their engagement and influence on the platform. The higher the SP, the greater the voting influence on content rewards. This mechanism encourages users to post high-quality content.

The distribution of post rewards depends on the SP of the voters. Each voter's SP deter-

mines their impact on the content's rewards. Voters with higher SP have a larger influence on the reward distribution. The final earnings of a post are not only determined by its quality but also by the number of votes and the SP of the voters. Voters share in the rewards for the post based on the proportion of their SP. Therefore, post earnings are closely related to the SP of the voters and their voting actions. Users with higher SP can not only bring more rewards to content creators but also earn corresponding rewards themselves.

4.2 Coalition Voting by Users on Posts

On the Steemit platform, users may join forces to vote on specific posts in order to receive more rewards, forming what is known as a user voting alliance. Steemit's reward mechanism is based on Steem Power (SP), and the voting weight is proportional to the amount of SP a user holds. Low-SP users can join a voting alliance, and by combining their voting power with that of other users, they can amplify the collective voting effect, thereby increasing the exposure and rewards of the post.

On one hand, voting alliances provide low-SP users with the opportunity to collaborate with other users, especially when there are strong social relationships between alliance members. The collective voting power becomes more significant in such cases. The social capital of alliance members, such as their number of followers, followings, and community activity duration, contributes to enhancing the voting effect, further boosting the post's reach. Users with more social capital can mobilize more supporters to participate in voting, thereby strengthening the overall effect of the voting alliance.

On the other hand, voting alliances increase the exposure of posts through concentrated voting, thereby boosting rewards. Voters also receive corresponding rewards based on their voting weight. Although this mechanism promotes the rewards for content creators and interaction within the platform, excessive concentrated voting may result in certain content receiving disproportionate rewards, affecting the diversity and fairness of the platform's content. Therefore, this study aims to examine the impact of alliance voting on post content.

5. Research Methodology

Our research process consists of four main stages, including data acquisition, calculation of coalition voting behavior variables using complex network analysis, construction of knowledge contribution variables, and application of the Panel Vector Autoregression (PVAR) model to explore the dynamic relationship between coalition voting and knowledge contribution.

5.1 Methodology and Data Description

This section outlines the research methodology and data utilized in our study, focusing on how complex network analysis and the PVAR model address the dynamic interactions between coalition voting and knowledge contribution. It also provides details on data sources, collection methods, and validation procedures, ensuring the robustness and reliability of our findings.

5.1.1 Data Collection and Preprocessing

Steemit, established in 2016, is the largest blockchain token-incentivized knowledge

community, with over 1.42 million registered users and a market circulation value of $70 million. Its unique token-based incentive mechanisms and transparent interaction records make it an ideal platform for exploring the relationship between voting behavior and knowledge contributions. The decentralized structure of Steemit, combined with its rich and accessible data, provides a robust foundation for studying user dynamics in a knowledge-sharing community.

We obtained the available data from the platform using Steemit's official API, ensuring the accuracy and authenticity of the extracted information.The dataset includes all activity records of community users in 2019, including voting data, post content, and social network connections. Since our PVAR model requires panel data, we ensured data completeness by selecting users who had posted at least once in each month from January to December 2019, resulting in a total of 5,674 users. This selection encompasses all eligible users in the complete dataset who meet the criteria. To address dimensional inconsistencies among variables, we standardized the data to ensure comparability across all variables in our PVAR model. After standardization, the data follows a standard normal distribution with a mean of 0 and a standard deviation of 1, aligning with the assumptions required for robust panel data analysis.

5.2 Construction of Coalitional Voting Variables

To comprehensively capture the complex dynamics of collective voting behavior, the coalitional voting score is derived through a multi-step process. This process not only relies on the frequency of joint voting occurrences but also incorporates techniques such as time filtering, historical voting overlap adjustment, and community detection. Specifically, time filtering excludes coincidental joint voting within short time windows, as multiple joint votes occurring in such brief periods are more likely to indicate deliberate coordination rather than random overlap. Historical voting overlap adjustment takes into account users' past voting activities, reducing the impact of random overlaps and ensuring that only genuine coalition behaviors are identified. Finally, community detection is used to construct a user network, applying the Louvain algorithm to identify long-term collaborative behaviors and potential voting coalitions. By combining these methods, we ensure the rigor and reliability of the coalitional voting score, effectively distinguishing genuine coalition behavior from coincidental overlaps driven by similar interests or random factors.

Firstly, we aggregate users' voting data across different posts and months to identify potential joint voting patterns. After aggregation, we filter joint voting behaviors that occur within long time windows to exclude coincidental overlaps that might result from similar interests. By analyzing time intervals, we can more accurately identify repeated joint voting that occurs within short periods, which often signals deliberate coordination (Rand et al. 2011). Ranjbar-Sahraei et al. (2014) explored cooperation in complex networks, emphasizing the relevance of time-sensitive interactions, further supporting the hypothesis of short time intervals. To filter out coinciden-

tal overlaps, we exclude joint voting behaviors that occur beyond a threshold defined as the mean time interval plus two standard deviations, which helps identify more deliberate coordination (Moore and McCabe 1989, Arnold 1990).

Secondly, we calculate the number of posts where users have jointly voted; however, this is not the final metric for coalitional voting assessment. Instead, we employ statistical methods to estimate the expected frequency of such joint voting behaviors. By computing the expected joint voting frequency (E_{uv}) based on each user's individual voting frequency and comparing it with the actual joint voting count (N_{uv}), we standardize these observations to differentiate genuine coalitional behavior from random coincidences.

To more rigorously define the concept of coalitions, we apply a standardized score (Z_{uv}) to quantify the deviation of observed joint voting behavior from the expected value in terms of standard deviations:

$$Z_{uv} = \frac{N_{uv} - E_{uv}}{\sqrt{E_{uv}}}$$

Only user pairs with $Z_{uv} > 3.0$ are considered likely to participate in coalitional voting behavior. This threshold filters out noise, ensuring that only significant anomalous voting behaviors, indicative of deliberate coalitions, are retained. This threshold selection is based on Fisher (1970)'s statistical methods for distinguishing valid voting behavior from random fluctuations.

Finally, based on the filtered results, we construct a user network. We reference existing research that shows complex network analysis is an effective tool for exploring collaborative behavior between users and forming voting coalitions (Tang et al. 2022, Li et al. 2021). Through this approach, we are able to identify non-random collaboration patterns among users, thus providing strong data support for analyzing coalition voting. We then apply the Louvain community detection algorithm (De Meo et al. 2011) to identify user groups that exhibit sustained collaborative behavior, which can be seen as potential voting coalitions. The Louvain algorithm is widely used in community detection due to its ability to efficiently detect community structures in large-scale networks. The strength of the Louvain algorithm lies in its modularity optimization method, which effectively handles large datasets and identifies stable community structures (Anuar et al. 2021). This enables us to accurately identify sustained collaborative behavior within user groups, providing an important theoretical and empirical basis for studying the formation mechanism of coalition voting. Each user's final coalition voting score is calculated by considering their participation in shared voting behaviors within these detected communities and the correlation of their voting activities' timing, further emphasizing consistent and non-random collaborative behavior.

5.3 Construction of Variable for Knowledge Contribution

Under a large extent, the level of knowledge sharing among community users determines the community's value and potential for long-term development (He and Wei 2009). Existing research primarily constructs indicators of user knowledge sharing based on either ques-

tionnaire design or actual user behavior data (Liao et al. 2020). Given that actual user behavior data is generally more persuasive than questionnaire surveys, and such data from the Steemit community is readily available, this study utilizes real data from the Steemit community to index user knowledge sharing. Knowledge contribution refers to the valuable content users provide in the community, encompassing both the number of posts and their innovation (Jin et al. 2015). For the Steemit community, the quality and innovation of community knowledge are incentivized goals, and user search results manifest as user voting behavior. Therefore, this study measures user knowledge contributions in two ways, which are the number of posts and knowledge innovation.

The number of posts refers to the quantity of content users publish, which directly reflects the level of user knowledge contribution in this study. We directly use the number of posts published by users during period t to represent this measure.

Knowledge innovation refers to the novel and creative content users provide in the community. To measure users' contributions to community knowledge innovation, this study constructs a knowledge co-occurrence network based on users' original articles over a certain period, selecting knowledge depth and breadth as the dimensions to describe knowledge innovation. Knowledge depth refers to the complexity of a user's knowledge, while knowledge breadth refers to the scope of a user's knowledge domain. First, we use the LDA topic model to predict the topic distribution of each article, considering topics with a probability exceeding a certain threshold as the knowledge elements of the article. Treating the knowledge elements in the article as nodes, we add an undirected edge between each co-occurring knowledge element to construct a knowledge co-occurrence network model. Second, we refer to expert knowledge representation methods based on knowledge networks, using the clustering coefficient of the knowledge symbiotic network as an index of user knowledge depth (Boh et al. 2014) and the network diameter as an index of user knowledge breadth (Boh et al. 2014) .

5.4 The Panel-VAR Methodology

In our analysis, we employ the Panel Vector Autoregression (PVAR) model developed by Love and Zicchino (Love and Zicchino 2006) to explore the dynamic relationship between coalition voting behavior and knowledge quality.

The PVAR model is widely applied across various fields, including economics, marketing, and knowledge management. It is particularly suitable for studying panel data with a large number of observations and a small number of time periods. This model allows us to control for unobserved individual heterogeneity, use lagged dependent variables as instruments within the Gaussian mixture model framework, and capture dynamic relationships based on users' past coalition voting behavior and user characteristics.

Our main model is the PVAR model, and its form is as follows: The endogenous variables are $\text{Number_of_Posts}_{u,i}$, $\text{Knowledge_Depth}_{u,i}$, $\text{Knowledge_Breadth}_{u,i}$, and $\text{Coalition_Voting}_{u,i}$. We also control for

exogenous variables, including $\text{Following}_{u,i}$, $\text{Followers}_{u,i}$, and $\text{Tenner}_{u,i}$.

$$\begin{bmatrix} \text{Number_of_Posts}_{u,i} \\ \text{Knowledge_Depth}_{u,i} \\ \text{Knowledge_Breadth}_{u,i} \\ \text{Coalition_Voting}_{u,i} \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} + \sum_{j=1}^{J} \begin{bmatrix} \beta_{1,1}^{j} & \cdots & \beta_{1,4}^{j} \\ \beta_{2,1}^{j} & \cdots & \beta_{2,4}^{j} \\ \beta_{3,1}^{j} & \cdots & \beta_{3,4}^{j} \\ \beta_{4,1}^{j} & \cdots & \beta_{4,4}^{j} \end{bmatrix} \begin{bmatrix} \text{Number_of_Posts}_{u,i-j} \\ \text{Knowledge_Depth}_{u,i-j} \\ \text{Knowledge_Breadth}_{u,i-j} \\ \text{Coalition_Voting}_{u,i-j} \end{bmatrix}$$
$$+ \begin{bmatrix} \gamma_{1,1} & \gamma_{1,2} & \gamma_{1,3} \\ \gamma_{2,1} & \gamma_{2,2} & \gamma_{2,3} \\ \gamma_{3,1} & \gamma_{3,2} & \gamma_{3,3} \\ \gamma_{4,1} & \gamma_{4,2} & \gamma_{4,3} \end{bmatrix} \begin{bmatrix} \text{Following}_{u,i} \\ \text{Followers}_{u,i} \\ \text{Tenure}_{u,i} \end{bmatrix} + \begin{bmatrix} \epsilon_{1,ui} \\ \epsilon_{2,ui} \\ \epsilon_{3,ui} \\ \epsilon_{4,ui} \end{bmatrix} \tag{1}$$

Where i is the month ($i = 1, 2, \cdots, 12$); u is the user; $\beta_{m,n}^{ij}$ and γ_{k}^{ij} are coefficients; α_{m}^{i} are constants representing panel fixed effects; $\epsilon_{m,ui}$ are idiosyncratic errors; j is the optimal lag length of the model. We log-transformed all endogenous variables to eliminate scaling effects and address concerns about heteroscedasticity.

6. Empirical Analysis

First, we determined the optimal lag length for the PVAR model. The optimal lag length refers to the ideal number of lags in the model, which can be evaluated using criteria such as the Modified Bayesian Information Criterion (MBIC), Modified Akaike Information Criterion (MAIC), and Modified Quasi Information Criterion (MQIC). The minimum value of each criterion determines the optimal lag length according to that particular criterion. The optimal lag length for the model is determined by the lag period with the smallest value across these three criteria. According to Table 1, we selected a lag length of 1.

Table 1 Model Selection Criteria

Lag	MAIC	MBIC	HQIC
1	11.3454	**14.9264***	**12.4606***
2	11.4202	15.3550	12.6516
3	**11.3219***	15.6926	12.6972
4	11.7781	16.7003	13.3366
5	12.6995	18.3429	14.4994

6.1 Generalized Method of Moments

The Granger causality test verifies whether a Granger causality exists between variables, but it does not verify the degree of interaction between variables. To address this, the Generalized Method of Moments (GMM) was employed to describe the results of the PVAR model estimation, which are reported in Table 2.

From the equation of coalition voting, the lagged variable *Following* does not have a statistically significant impact on the current coalition voting ($\alpha = -0.0910$, $p > 0.1$). Therefore, the hypothesis H1 is not supported. A possible reason is that coalition voting may not be directly influenced by the number of users a participant follows but rather by other

Table 2 Regression Results

VARIABLES	Post_count	Knowledge_depth	Knowledge_breadth	Coalition_voting
L.Post_count	0.699***	0.107***	0.0991***	0.0257***
	(0.0216)	(0.00774)	(0.00856)	(0.00583)
L.Knowledge_depth	0.0671***	0.413***	−0.0304	0.324***
	(0.0114)	(0.0125)	(0.0185)	(0.0132)
L.Knowledge_breadth	−0.0357***	−0.146***	−0.0870***	0.0215*
	(0.00714)	(0.00990)	(0.0120)	(0.0121)
L.Coalition_voting	0.00402	0.102***	0.0928***	0.350***
	(0.00574)	(0.0121)	(0.0127)	(0.0431)
L.Following	−0.0833***	−0.112***	−0.0151	0.101***
	(0.0197)	(0.0267)	(0.0165)	(0.0291)
L.Follower	−0.0232	−0.00586	−0.00158	−0.0910
	(0.0258)	(0.0218)	(0.0235)	(0.0820)
L.Tenure	−0.000991	−0.0149	0.0295	−0.203***
	(0.0185)	(0.0165)	(0.0201)	(0.0360)

Notes: *** p<0.01, ** p<0.05, * p<0.1

network or contextual factors. The lagged variable *Follower* also does not have a statistically significant impact on coalition voting ($\alpha = 0.101$, $p < 0.01$). Hence, the hypothesis H2 is supported. Additionally, tenure has a significant positive impact on coalition voting ($\alpha = -0.203$, $p < 0.01$). This suggests that users who joined the community earlier are more likely to participate in coalition voting, indicating that early adopters tend to have greater involvement in collaborative activities.. Thus, the hypothesis H3 is supported.

The lagged one-period coalition voting ($\alpha = 0.0257$, $p < 0.01$) has a significant positive impact on the current post count, indicating that users who participated in coalition voting in the past are more likely to post in the current period. Hypotheses H4 *are supported*. This may be because users who engage in coalition voting are more active in the community and more willing to post content to gain more attention and votes. Lagged coalition voting ($\alpha = 0.102$, $p < 0.01$) shows a positive impact on knowledge depth, indicating that participation in coalition voting contributes to the improvement of users' knowledge depth. Therefore, hypothesis H5 is supported. Finally, lagged coalition voting ($\alpha = 0.0928$, $p < 0.01$) has a significant positive effect on knowledge breadth, suggesting that participation in coalition voting facilitates expanding users' areas of knowledge. Hence, hypothesis H6 is supported.

6.2 Granger Causality Tests

We conducted the Granger causality tests on our results to gain insights into the causal relationships between the variables. As shown in Table 3,coalition voting significantly influenced the number of user posts, the depth of user knowledge, and the breadth of user knowledge. These results are consistent with

the GMM analysis, confirming that users who engage in coalition voting are more likely to actively participate by posting more content and expanding both their knowledge breadth and depth. Therefore, hypotheses H4, H5, and H6 are supported.

Table 3 Granger Causality Tests

Excluded → Equation	χ^2
CV → Post Count	19.415***
CV → Knowledge Depth	2.7007*
CV → Knowledge Breadth	3.1494*
CV → ALL	165.8***

6.3 Impulse Response Functions

The impulse response function can be used to analyze the effect of shocks on a variable over different periods through a time window, thereby exploring the interaction and degree of influence between variables. In this study, the impulse response function serves as additional evidence for our hypotheses, illustrating the dynamic nature of the mutual influences between voting behavior and user knowledge contribution, particularly when the impact reaches its highest and lowest points.

Figure 1 shows the impulse response function results, illustrating the impact of coalition voting on itself, knowledge breadth, knowledge depth, and post count. In Figure 1(a), the response of coalition voting to its own shock starts at approximately 0.51 and gradually declines to zero by the 10th month. This smooth decay indicates that the effect of the initial shock diminishes steadily over time.

In Figure 1(b), knowledge depth responds positively to the coalition voting shock, rising to around 0.023 by the second month before gradually approaching zero by the 10th

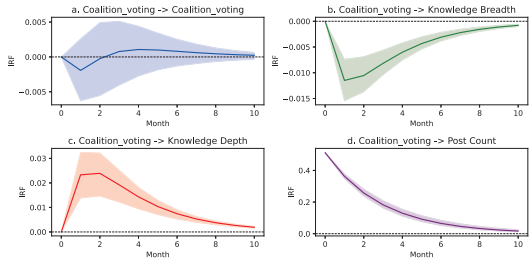

Figure 1 IRF Results

month. This suggests that coalition voting has a short-term positive effect on knowledge depth, which fades over time.

In Figure 1(c), the response of knowledge breadth is initially slightly negative, reaching about -0.011 in the second month before returning to zero by the 10th month. This indicates a minor, temporary reduction in knowledge breadth due to coalition voting, which ultimately dissipates.

In Figure 1(d), post count initially increases in response to the coalition voting shock, starting at approximately 0.36, then steadily declines towards zero by the 10th month. This suggests that coalition voting has a positive but short-lived impact on user activity in terms of posting.

6.4 Robustness Check

We conducted additional tests to verify the robustness of our results. Using non-stationary data in regression analysis can lead to spurious regression, thereby undermining the reliability of the model estimation results. In this study, we employed the Levin, Lin, and Chu (LLC) test to assess the stationarity of the relevant variables in each data group. The results confirmed that all variables passed the unit root test, as shown in Table 4.

Additionally, we performed the eigenvalue stability test for the PVAR model to check

Table 4 Levin-Lin-Chu Unit-Root Test

Variable	Adjusted t*
Post Count	−2.6e+02***
Knowledge Depth	−1.1e+02***
Knowledge Breadth	−1.5e+02***
Coalition Voting	−43.7819***
Following	−23.0983***
Followers	−46.4440***

Note: *, **, and *** denote significance at the 10%, 5%, and 1% levels.

whether all the eigenvalues lie within the unit circle. The results, as shown in Figure 2, indicate that all the eigenvalues have a modulus less than 1, confirming that our model is stable.

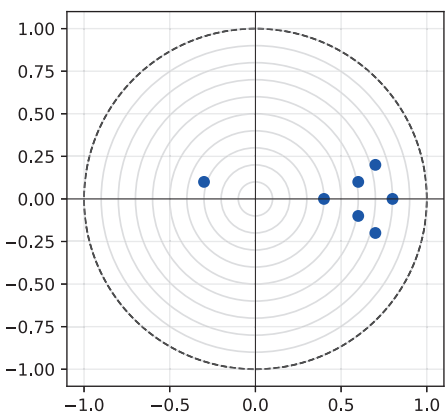

Figure 2 Eigenvalue Stability

7. Discussion and Conclusion

This paper investigates the impact of user coalition voting on their knowledge contribution in blockchain platforms. We analyzed the relationship between user characteristics and coalition voting and how coalition voting influences users' knowledge contribution behaviors in the community. By using a rich dataset from Steemit and various estimation methods, we found that user-related factors (including the number of users followed and the duration of community membership) are related to user participation in coalition voting. Specifically, users who follow more people and have a shorter duration of membership are more likely to join coalition voting. However, the number of followers a user has does not have much direct impact on coalition voting. Additionally, we explored the impact of user coalition voting on their knowledge contribution behaviors. On the one hand, coalition voting can have a positive impact on the number of posts, as well as the breadth and depth of knowledge in user posts.

7.1 Conclusion

The empirical results provide a comprehensive insight into the impact of coalition voting on knowledge contributions within the Steemit community. Contrary to H1, which posited that users with a higher number of followers are more likely to participate in coalition voting, the analysis revealed that the number of followers does not have a significant effect on coalition voting behavior. This unexpected result suggests that merely having a large number of followers does not necessarily lead to higher participation in coalition voting activities.

Conversely, H2 was confirmed, with results indicating that users with a higher number of followings are more likely to engage in coalition voting. Specifically, the number of followings has a significant positive effect on coalition voting, suggesting that individuals who follow more users are more inclined to participate in collaborative voting activities. This may be because users with extensive follow relationships

possess stronger social capital and resources within the community, enabling them to more effectively organize and mobilize coalition voting activities (Swanson et al. 2020, Wang et al. 2024).

Additionally, H3 was also supported, showing that users with shorter community membership durations are more likely to engage in coalition voting. This finding is consistent with Knoben (2008), which posits that new users seek to rapidly establish their reputation and influence through active participation and organization of coalition voting.

Notably, H4, H5, and H6 were all confirmed, indicating that participation in coalition voting significantly enhances users' knowledge contributions, including the number of posts, knowledge depth, and knowledge breadth. Specifically, coalition voting behavior is positively correlated with the number of posts, suggesting that users who participate in coalition voting are more active and willing to publish content to gain more attention and rewards. Furthermore, the increases in knowledge depth and breadth indicate that coalition voting facilitates users' exploration and specialization in different knowledge areas, thereby enriching the overall knowledge structure of the community. These results emphasize that coalition voting not only promotes effective governance but also enhances users' substantive contributions, thereby driving the sustained vitality and innovation within Decentralized Autonomous Organizations (DAOs).

7.2 Implications

First, from a theoretical perspective, this study enriches the literature on knowledge management in decentralized platforms by revealing the role of coalition voting in promoting knowledge contribution. Traditional studies have primarily focused on individual incentives and their direct impact on knowledge sharing (Bock et al. 2005, Chen et al. 2019). In contrast, this study highlights the role of collective behavior, particularly coalition voting, in enhancing both the quantity and quality of knowledge contributions. This dual impact emphasizes the importance of considering group dynamics and collaborative strategies in knowledge management models within blockchain communities.

Second, the study finds a positive correlation between coalition voting and knowledge innovation (including knowledge depth and breadth), which further deepens our understanding of how cooperative behavior promotes more specialized and diversified knowledge production. This is consistent with social network theory, which posits that diverse and interconnected networks facilitate the flow of new information (Granovetter 1973). This study extends this theory to the context of DAOs, demonstrating that coalition voting not only influences governance outcomes but also enriches the intellectual capital of the community.

From a practical perspective, this study provides actionable insights for DAO communities in balancing effective governance with robust knowledge sharing. The positive impact of coalition voting on user engagement and knowledge contribution suggests that pro-

moting collaborative voting behavior can enhance community vitality. DAO administrators can consider designing governance mechanisms that encourage the formation of coalitions, thereby leveraging collective efforts to drive decision-making and knowledge dissemination. Additionally, the coalition voting detection algorithm developed in this study can be integrated into the DAO's governance framework to monitor and regulate collective behavior, ensuring that it contributes positively to community goals without undermining fairness and justice.

7.3 Limitations and Future Studies

This study has several limitations that future research can address. First, coalition voting in this paper is based on users' long-term voting behaviors. Future research should distinguish between different coalitions to explore their impacts on the community. Second, this paper only discusses the impact of coalition voting on knowledge contribution. Future research could delve into other aspects, such as knowledge interaction and knowledge search. Finally, the PVAR model is a data-driven method that can describe the time-varying and feedback relationships between variables in a closed system. Future research can use methods like system dynamics simulation to study the dynamic model of a knowledge community ecosystem with complex blockchain token incentives.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grant Nos. 72071083 and 72171089; the National Social Science Fund of China under Grant No. 23ZDA063; and the Fundamental Research Funds for the Central Universities under Grant Nos. 2023ZYGXZR021 and ZDPY202304.

The authors sincerely thank the anonymous reviewers for their valuable feedback, which greatly improved the quality of this paper, and the members of the editorial board for considering our manuscript for possible publication.

Data Availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Conflict of Interest

The authors declare no conflict of interest.

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